Quantium Task 2 - Trial vs Control Store Analysis

```
# Imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv("E:/Work/Forage/Quantium/Task 2/QVI_data.csv")
df.head(5)
df['DATE']=pd.to_datetime(df['DATE'])
df['YEARMONTH']=df['DATE'].dt.strftime('%Y%m')
```

Monthly Store Metrics

```
df1 = df.groupby(['STORE_NBR','YEARMONTH'],as_index=False).agg(
    total_sales=('TOT_SALES','sum'),
    unique_cust=('LYLTY_CARD_NBR','nunique'),
    unique_txn=('TXN_ID','nunique')
)

df1['trans_per_cust'] = df1['unique_txn'] / df1['unique_cust']
df1.head()
```

	STORE_NBR	YEARMONTH	total_sales	unique_cust	unique_txn	trans_per_cust
0	1	201807	206.9	49	52	1.061224
1	1	201808	176.1	42	43	1.023810
2	1	201809	278.8	59	62	1.050847
3	1	201810	188.1	44	45	1.022727
4	1	201811	192.6	46	47	1.021739

Pre-Trial Period

```
pre_trial_months = [
    '201807','201808','201809',
    '201810','201811','201812',
    '201901'
]
```

```
df_pre_trial = df1[df1['YEARMONTH'].isin(pre_trial_months)]
```

Keep Stores with Complete History

```
store_month_count = df_pre_trial.groupby('STORE_NBR')
['YEARMONTH'].count()

valid_stores = store_month_count[store_month_count == 7].index

df_pre_trial_complete = df_pre_trial[
    df_pre_trial['STORE_NBR'].isin(valid_stores)
]
```

Sales Pivot Table

```
df_pivot_sales = df_pre_trial_complete.pivot(
    index='YEARMONTH',
    columns='STORE_NBR',
    values='total_sales'
)

df_pivot_sales.head()
```

```
STORE_NBR    1      2        3       4      5      6       7      8
9    \
YEARMONTH

201807     206.9  150.8  1205.70  1399.9  812.0  260.0  1024.7  381.6
289.7
201808     176.1  193.8  1079.75  1259.5  745.1  203.2  1119.9  383.0
327.5
201809     278.8  154.4  1021.50  1198.6  896.0  207.7  1147.3  293.0
369.7
201810     188.1  167.8  1037.90  1346.4  798.0  292.4  1063.9  422.8
361.0
201811     192.6  162.9  1008.00  1212.0  771.4  255.3  1076.5  349.2
374.4

STORE_NBR      10   ...   263    264    265    266   267     268
269  \
YEARMONTH           ...

201807     892.00  ...  38.7  232.6  247.8  127.3   6.2  224.00
982.0
201808     878.65  ...  28.0  203.3  227.1  154.5  24.9  322.65
835.1
201809     945.00  ...  21.8  199.5  133.5  139.0  23.3  174.40
886.0
201810     910.80  ...  36.6  170.0  162.9  170.4  14.4  237.60
```

```
1078.4
201811     885.00  ...  15.8  184.5  282.8  144.2  40.3  225.40
967.2

STORE_NBR       270    271     272
YEARMONTH
201807       962.80  956.6  433.10
201808      1003.75  683.9  372.85
201809       845.40  798.4  304.70
201810       816.40  790.0  430.60
201811       965.00  886.4  376.20

[5 rows x 260 columns]
```

Function to Find Control Stores

```
def find_control_store(store_nbr):
    return (
        df_pivot_sales
        .corrwith(df_pivot_sales[store_nbr])
        .sort_values(ascending=False)
        .head(10)
    )

find_control_store(77)
```

```
STORE_NBR
77     1.000000
71     0.914106
233    0.903774
119    0.867664
17     0.842668
3      0.806644
41     0.783232
50     0.763866
157    0.735893
162    0.729740
dtype: float64
```

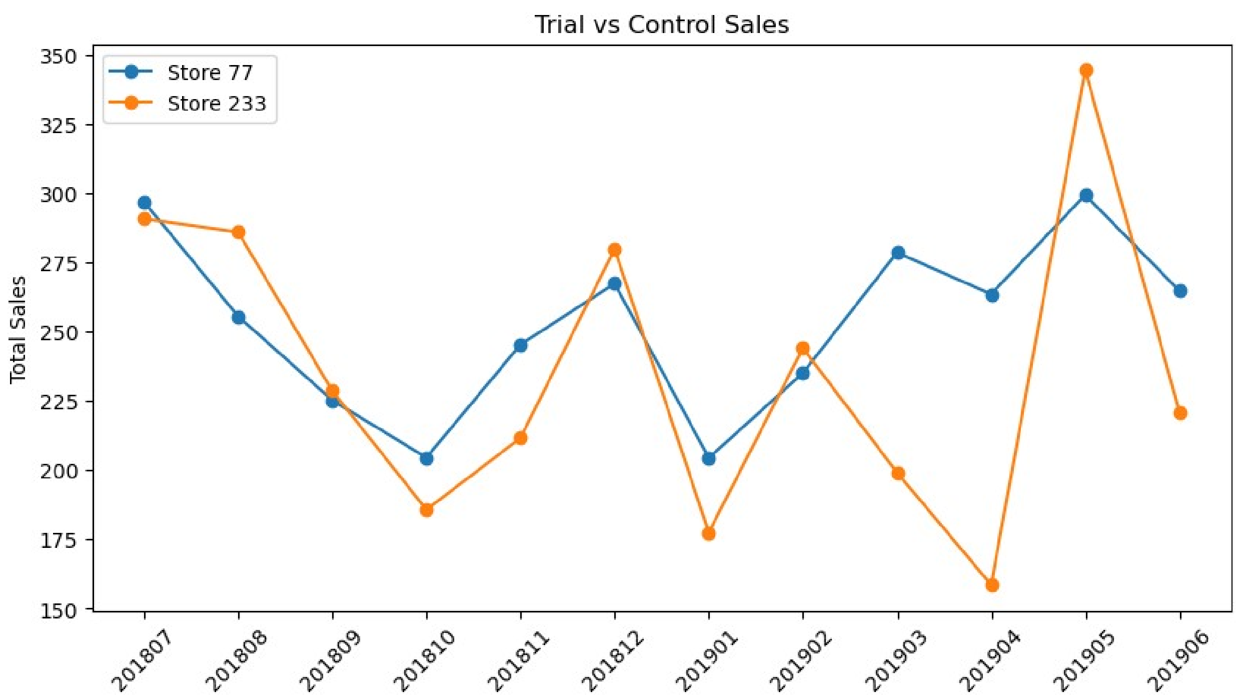
Selected Trial-Control Pairs

- 77 → 233
- 86 → 155
- 88 → 40

```
trial_control_pairs = {
    77:233,
    86:155,
    88:40
}
```

Sales Trend Plot

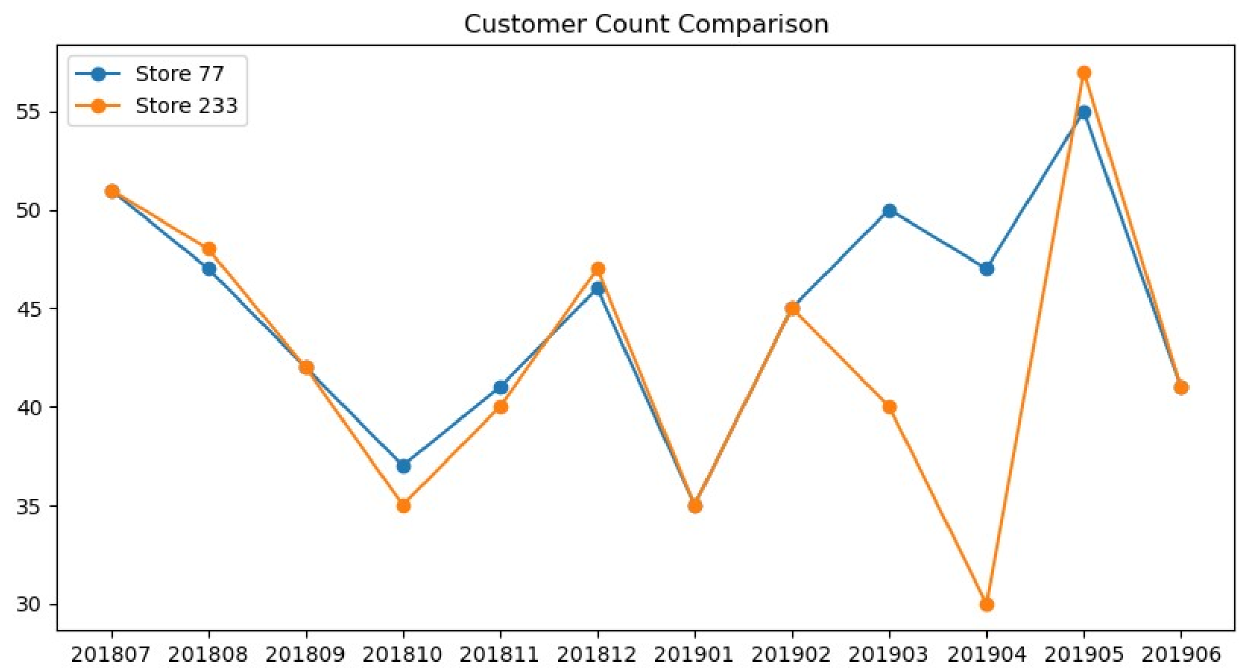
```
trial_store = 77
control_store = 233

plot_df = df1[df1['STORE_NBR'].isin([trial_store,control_store])]

plt.figure(figsize=(10,5))

for store in [trial_store,control_store]:
    temp = plot_df[plot_df['STORE_NBR']==store]
    plt.plot(temp['YEARMONTH'],
             temp['total_sales'],
             marker='o',
             label=f'Store {store}')

plt.title('Trial vs Control Sales')
plt.ylabel('Total Sales')
plt.xticks(rotation=45)
plt.legend()
plt.show()
```



Customer Count Trend

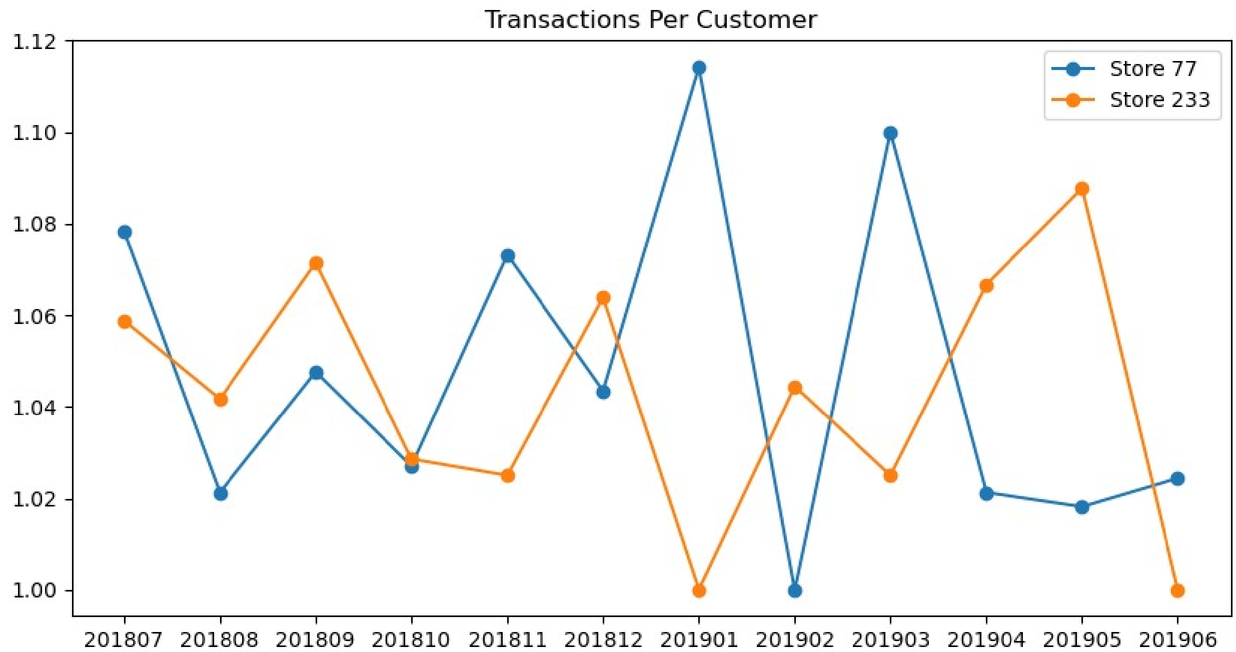
```
plt.figure(figsize=(10,5))

for store in [trial_store,control_store]:
    temp = plot_df[plot_df['STORE_NBR']==store]
```

```
    plt.plot(temp['YEARMONTH'],
             temp['unique_cust'],
             marker='o',
             label=f'Store {store}')

plt.title('Customer Count Comparison')
plt.legend()
plt.show()
```



Transactions Per Customer

```
plt.figure(figsize=(10,5))

for store in [trial_store,control_store]:
    temp = plot_df[plot_df['STORE_NBR']==store]
    plt.plot(temp['YEARMONTH'],
             temp['trans_per_cust'],
             marker='o',
             label=f'Store {store}')

plt.title('Transactions Per Customer')
plt.legend()
plt.show()
```




```
trial_months = ['201902','201903','201904']

df_trial = df1[df1['YEARMONTH'].isin(trial_months)]
df_trial.head()
```

```
    STORE_NBR YEARMONTH  total_sales  unique_cust  unique_txn
trans_per_cust
7           1    201902        225.4           52          55
1.057692
8           1    201903        192.9           45          49
1.088889
9           1    201904        192.9           42          43
1.023810
19          2    201902        139.4           29          32
1.103448
20          2    201903        192.1           43          46
1.069767
```

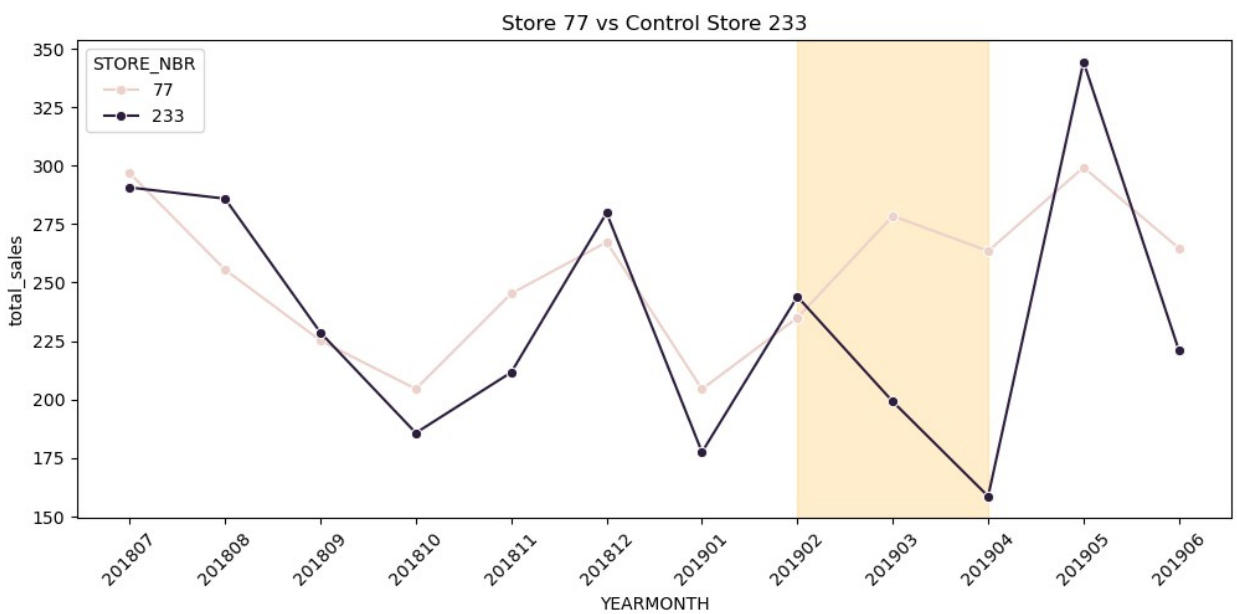
```
trial_control_pairs = {
    77:233,
    86:155,
    88:40
}

results = []

for trial, control in trial_control_pairs.items():

    trial_sales = df_trial.loc[
        df_trial['STORE_NBR']==trial,
```

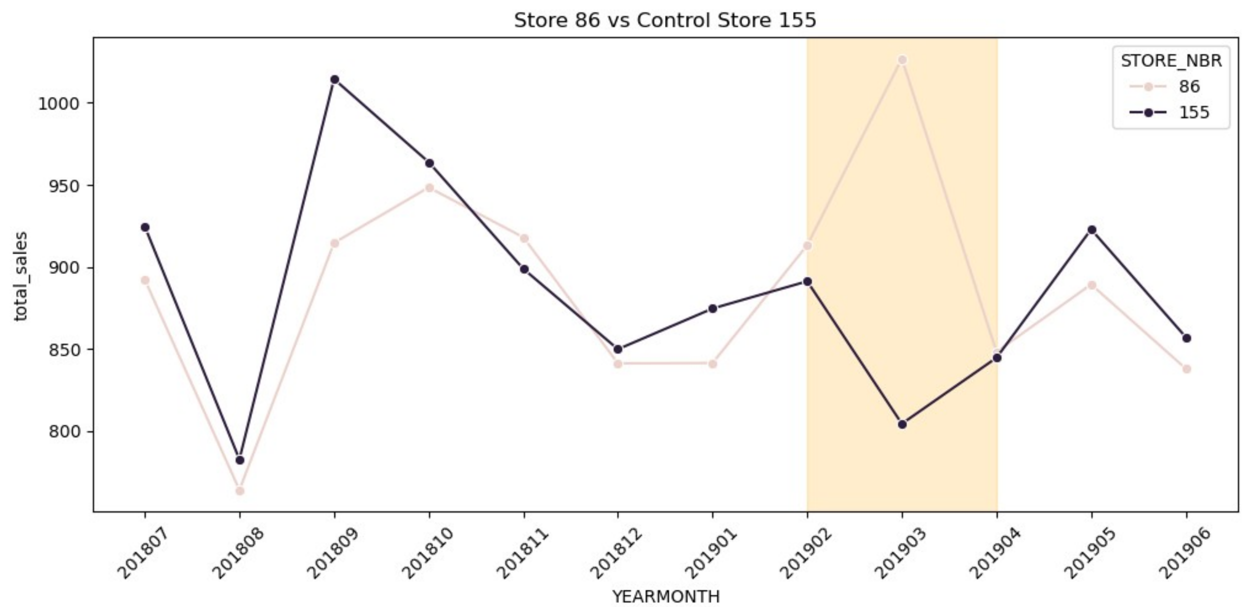
```
        'total_sales'
    ].sum()

    control_sales = df_trial.loc[
        df_trial['STORE_NBR']==control,
        'total_sales'
    ].sum()

    pct_diff = (
        (trial_sales-control_sales)
        / control_sales
    ) * 100

    results.append([
        trial,
        control,
        trial_sales,
        control_sales,
        pct_diff
    ])

sales_comparison = pd.DataFrame(
    results,
    columns=[
        'Trial Store',
        'Control Store',
        'Trial Sales',
        'Control Sales',
        '% Difference'
    ]
)

sales_comparison
```

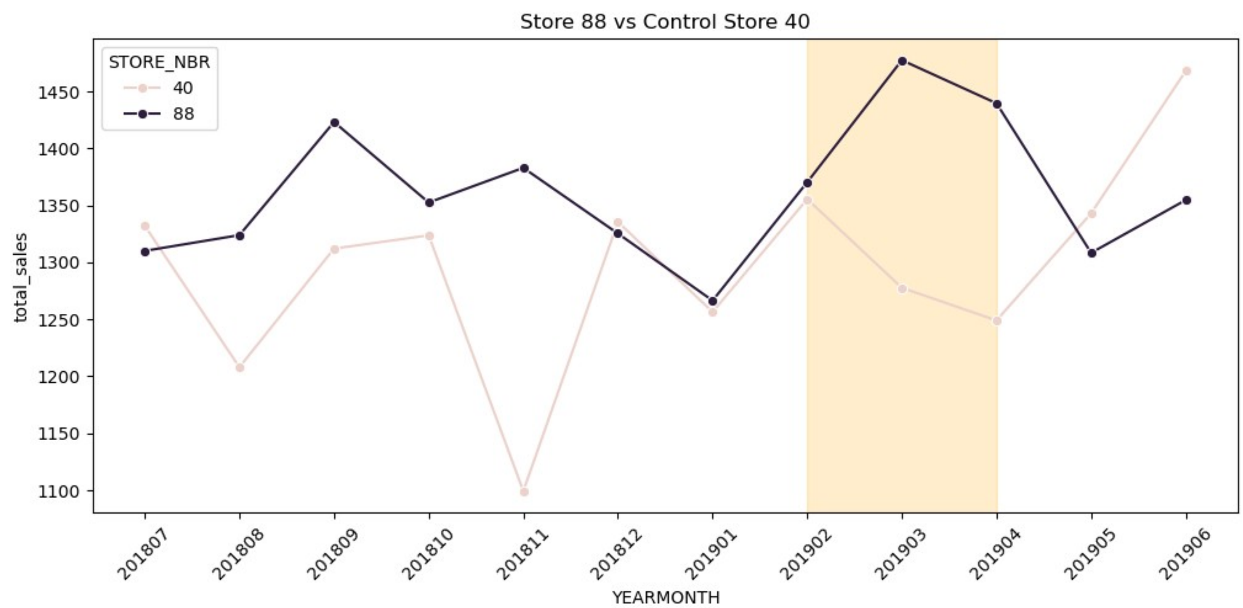
	Trial Store	Control Store	Trial Sales	Control Sales	% Difference
0	77	233	777.0	601.7	29.134120
1	86	155	2788.2	2540.2	9.763011
2	88	40	4286.8	3882.0	10.427615

```
for trial, control in trial_control_pairs.items():

    temp = df1[
        df1['STORE_NBR'].isin([trial,control])
    ]

    plt.figure(figsize=(12,5))
```

```
sns.lineplot(
    data=temp,
    x='YEARMONTH',
    y='total_sales',
    hue='STORE_NBR',
    marker='o'
)

plt.axvspan(
    '201902',
    '201904',
    alpha=0.2,
    color='orange'
)

plt.title(
    f'Store {trial} vs Control Store {control}'
)

plt.xticks(rotation=45)
plt.show()
```








```
customer_results = []

for trial, control in trial_control_pairs.items():

    trial_customers = df_trial.loc[
        df_trial['STORE_NBR']==trial,
        'unique_cust'
    ].sum()

    control_customers = df_trial.loc[
        df_trial['STORE_NBR']==control,
        'unique_cust'
```

```
    ].sum()

    pct_diff = (
        (trial_customers-control_customers)
        / control_customers
    ) * 100

    customer_results.append([
        trial,
        control,
        trial_customers,
        control_customers,
        pct_diff
    ])

customer_comparison = pd.DataFrame(
    customer_results,
    columns=[
        'Trial Store',
        'Control Store',
        'Trial Customers',
        'Control Customers',
        '% Difference'
    ]
)

customer_comparison
```

```
   Trial Store  Control Store  Trial Customers  Control Customers  \
0           77            233              142                115
1           86            155              327                288
2           88             40              386                362

   % Difference
0     23.478261
1     13.541667
2      6.629834
```

```
txn_results = []

for trial, control in trial_control_pairs.items():

    trial_txn = df_trial.loc[
        df_trial['STORE_NBR']==trial,
        'trans_per_cust'
    ].mean()

    control_txn = df_trial.loc[
        df_trial['STORE_NBR']==control,
        'trans_per_cust'
```

```
    ].mean()

    txn_results.append([
        trial,
        control,
        trial_txn,
        control_txn
    ])

txn_comparison = pd.DataFrame(
    txn_results,
    columns=[
        'Trial Store',
        'Control Store',
        'Trial Txn/Customer',
        'Control Txn/Customer'
    ]
)

txn_comparison

   Trial Store  Control Store  Trial Txn/Customer  Control
Txn/Customer
0           77            233            1.040426
1.045370
1           86            155            1.235704
1.261077
2           88             40            1.253563
1.187795
```

Business Conclusions

Trial Store Assessment

Control stores were selected based on similarity in pre-trial sales performance, customer counts and transactions per customer. The selected control stores were Store 233 for Trial Store 77, Store 155 for Trial Store 86 and Store 40 for Trial Store 88.

Store 77 delivered the strongest sales uplift during the trial period, outperforming its control store by 29.1%. The improvement was largely driven by a 23.5% increase in customer numbers while purchasing frequency remained stable.

Store 86 outperformed its control store by 9.8%. Customer counts increased by 13.5%, indicating that the trial was effective in attracting additional shoppers despite a small decline in transactions per customer.

Store 88 achieved a 10.4% sales uplift relative to its control store. Growth was driven by both higher customer numbers and increased transactions per customer, indicating strong customer engagement during the trial.

Overall, all three trial stores outperformed their respective control stores during the trial period. The strongest performance was observed in Store 77, while Store 88 demonstrated the most balanced improvement across both customer acquisition and purchasing behaviour.

Recommendation: The trial should be considered successful and the initiative should be evaluated for rollout to similar stores across the network.